Chemical and Electrochemical Stability Analysis of Fe-Halide Cathodes with Halide Solid Electrolytes

Computational Methodology and Results Summary

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1 Project Overview

This report documents the computational analysis of chemical and electrochemical stability at the interface between iron-halide cathode materials and halide solid electrolytes (SEs). The study targets three cathode compositions:

- **Row 3**: FeF_3 cathode vs. five halide SEs
- **Row 4**: $FeCl_3$ cathode vs. five halide SEs
- **Row 2**: FeF_3+$FeCl_3$ (1:1 molar) cathode vs. five halide SEs

The five halide solid electrolytes studied are:

Label	Formula
Li_3YCl_6	Li_3YCl_6
$Li_{2.4}Y_{0.4}Zr_{0.6}Cl_6$	$Li_{2.4}Y_{0.4}Zr_{0.6}Cl_6$
Li_3ScCl_6	Li_3ScCl_6
$Li_{2.375}Sc_{0.375}Zr_{0.625}Cl_6$	$Li_{2.375}Sc_{0.375}Zr_{0.625}Cl_6$
Li_3InCl_6	Li_3InCl_6

All thermodynamic data are retrieved from the Materials Project (MP) database via the `mp-api` client and processed using `pymatgen`.

2 Theoretical Framework

2.1 Chemical Stability (Closed System)

The chemical reaction energy at a mixing ratio x (molar fraction of cathode) in the closed system is:

$$\Delta E_{\text{rxn}}(x) = E^{\text{PD}}_{\text{hull}}[c(x)] - \left[x \cdot E^{\text{PD}}_{\text{hull}}(C) + (1-x) \cdot E^{\text{PD}}_{\text{hull}}(SE)\right] \tag{1}$$

where $c(x) = x\,C + (1-x)\,SE$ is the mixed composition, and $E^{\text{PD}}_{\text{hull}}$ denotes the energy on the convex hull of the standard `PhaseDiagram` (per atom, relative to the stable phases in the relevant chemical subsystem). A negative ΔE_{rxn} means the interface is thermodynamically unstable and will spontaneously react.

2.2 Electrochemical Stability (Open Li System)

In a battery, lithium can flow into or out of the interface region. This is modeled using the *grand potential* phase diagram (GPCD), in which Li is treated as an open element with chemical potential

$$\mu_{\text{Li}}(V) = \mu^{\text{ref}}_{\text{Li}} - V \tag{2}$$

where V is the electrode voltage (V vs. Li/Li^+) and $\mu^{\text{ref}}_{\text{Li}}$ is the Li chemical potential at the most Li-rich transition point on the phase diagram:

```
mu_li_ref = pd_obj.get_transition_chempots(Element("Li"))[0]
chempots = {"Li": -voltage + mu_li_ref}
gpd = GrandPotentialPhaseDiagram(entries, chempots)
```

The grand-potential energy of a composition c is:

$$\Phi(c) = E_{\mathrm{DFT}}(c) - n_{\mathrm{Li}}(c)\,\mu_{\mathrm{Li}} \tag{3}$$

The electrochemical reaction energy is then computed analogously to Eq. (1) but using Φ evaluated on the GPCD hull instead of E on the PD hull.

Physical interpretation. At *high voltage* (oxidizing conditions), the Li chemical potential is very negative ($\mu_{\mathrm{Li}} \ll 0$), so it costs energy to reduce Li from the interface; the cathode and SE are more stable (less negative ΔE_{rxn}). At *low voltage* (reducing conditions), Li is readily inserted, driving strong reduction reactions and more negative ΔE_{rxn}. This is why the low-voltage (blue) curves in the plots sit at the most negative energies.

2.3 Voltage Range and Normalization

- Voltage sweep: 1.0–4.2 V vs. Li/Li^+, step 0.01 V (321 steps)
- Energy normalization: eV per atom of the reactant *mixture* (not per formula unit); this is "`norm=True`" in `InterfacialReactivity`.
- x axis: molar fraction of cathode, $x = \frac{n_C/N_C}{n_C/N_C + n_{SE}/N_{SE}}$, where n_C, n_{SE} are the molar amounts and N_C, N_{SE} are the number of atoms per formula unit.

3 Rows 3 and 4: Binary Interface Calculations

3.1 Method: `InterfacialReactivity`

For the single-cathode cases (FeF_3 or $FeCl_3$ vs. each SE), pymatgen's built-in `InterfacialReactivity` and `GrandPotentialInterfacialReactivity` classes are used directly. These classes scan the tie-line between two compositions in the (grand potential) phase diagram, locate the "kink" points where the set of stable decomposition products changes, and return the reaction energy at each kink.

```
# Chemical stability (closed system):
iface = InterfacialReactivity(
    comp1, comp2, pd_obj,
    norm=True, use_hull_energy=False)

# Electrochemical stability (open Li, voltage V):
chempots = {"Li": -voltage + mu_li_ref}
gpd = GrandPotentialPhaseDiagram(entries, chempots)
iface = GrandPotentialInterfacialReactivity(
    comp1, comp2, gpd,
    pd_non_grand=pd_obj,
    include_no_mixing_energy=True,
    norm=True, use_hull_energy=False)

for _, x_atom, energy, rxn, _kJ in iface.get_kinks():
    x_molar = atom_to_molar(x_atom, n1, n2)
    ...
```

The atom-fraction output from `get_kinks()` is converted to molar fraction via:

$$x_{\text{mol}} = \frac{x_{\text{atom}}/N_C}{x_{\text{atom}}/N_C + (1 - x_{\text{atom}})/N_{SE}} \tag{4}$$

3.2 Results: FeF_3 vs. Halide SEs (Row 3)

Outputs: `chemical_stability_FeF3.png`, `electrochemical_stability_FeF3.png`, `stability_table_FeF3.csv` (652 non-zero-energy rows).

Key observations:

- FeF_3 shows a clear phase boundary at approximately $x \approx 0.57$ for the Cl-based SEs, corresponding to a mixed Fe–F–Cl–Li product ($LiYF_4$ + LiCl + Fe metal).
- Reaction energies at $V = 1\,\text{V}$ range from -0.65 to $-1.36\,\text{eV/atom}$ depending on the SE, indicating significant thermodynamic instability at low voltages.
- At $V = 4.2\,\text{V}$ the energies approach ~ -0.1 to $-0.3\,\text{eV/atom}$, showing improved but non-negligible instability even at high voltage.
- The Zr-substituted electrolytes ($Li_{2.4}Y_{0.4}Zr_{0.6}Cl_6$, $Li_{2.375}Sc_{0.375}Zr_{0.625}Cl_6$) show additional intermediate kinks due to the larger element space (F, Cl, Zr present).

3.3 Results: $FeCl_3$ vs. Halide SEs (Row 4)

Outputs: `chemical_stability_FeCl3.png`, `electrochemical_stability_FeCl3.png`, `stability_table_FeCl3.c` (282 non-zero-energy rows).

Key observations:

- $FeCl_3$ shows a simpler reaction landscape than FeF_3: in most cases only a single interior kink exists (at $x = 1$, i.e., pure cathode), corresponding to $FeCl_3 + 3\,Li \rightarrow Fe + 3\,LiCl$.
- At $V = 1\,\text{V}$, $\Delta E_{\text{rxn}} \approx -1.29\,\text{eV/atom}$, substantially negative.
- The energy decreases nearly linearly with increasing voltage, reaching $\approx -0.07\,\text{eV/atom}$ near $4.2\,\text{V}$, indicating $FeCl_3$ is relatively stable at high operating voltages.
- Because F is absent in the $FeCl_3$ system, no LiF or mixed fluoride products form, leading to fewer reaction channels.

3.4 Database Note: Li_3YCl_6

Li_3YCl_6 is not stable on the convex hull in the current MP database (version 2025.09.25). Pymatgen issues a warning and falls back to the hull energy, which corresponds to the decomposition $3\,LiCl + YCl_3$. This is physically reasonable and was accepted as-is. Results for the Y-containing SEs therefore reflect this projected hull energy rather than the energy of a single Li_3YCl_6 phase.

4 Row 2: Mixed Cathode FeF_3+$FeCl_3$ (1:1 Molar)

Two approaches were implemented for the mixed cathode.

4.1 Option A: Pseudo-Compound Approximation

4.1.1 Method

The 1:1 molar mixture of FeF_3 and $FeCl_3$ is combined into a single pseudo-compound:

$$1\,\text{mol FeF}_3 + 1\,\text{mol FeCl}_3 \longrightarrow \text{Fe}_2\text{F}_3\text{Cl}_3 \tag{5}$$

This composition $Fe_2F_3Cl_3$ is then used as `comp1` in the same `GrandPotentialInterfacialReactivity` pipeline as Rows 3 and 4. $Fe_2F_3Cl_3$ is not a stable MP entry; pymatgen projects it onto the convex hull of the relevant chemical system (Fe, F, Cl, + SE elements + Li). In the reaction strings it appears as $Fe_2(ClF)_3$.

4.1.2 Results

Outputs: `chemical_stability_FeF3-FeCl3.png`, `electrochemical_stability_FeF3-FeCl3.png`, `stability_table_FeF3-FeCl3.csv` (687 rows).

- A single dominant kink at $x \approx 0.57$ across all SEs, similar to the FeF_3 case (same element space, same phase space projection).
- At $V = 1\,\text{V}$ vs. Li_3YCl_6: $\Delta E_{\text{rxn}} \approx -0.84\,\text{eV/atom}$ at $x = 0.57$ and $-1.33\,\text{eV/atom}$ at $x = 1$ (pure cathode boundary), intermediate between the pure FeF_3 (-0.65, -1.36) and $FeCl_3$ (-1.29) values.
- The mixing of F and Cl in the pseudo-compound leads to $LiYF_4$ + LiCl + Fe as the primary products at the intermediate kink.

4.1.3 Limitations

Because $Fe_2F_3Cl_3$ is treated as a *single entity*, the reference energy on the cathode side is the hull energy of $Fe_2F_3Cl_3$ (projected). If FeF_3 and $FeCl_3$ can independently and separately react with the SE (different products at different spatial locations), Option A does not capture this. Nor does it capture any intrinsic chemical reactivity *between* FeF_3 and $FeCl_3$ themselves.

4.2 Option C: Ternary Grand-Potential Approach

4.2.1 Motivation

In reality the cathode is a *physical mixture* of two separate phases, not a single compound. Option C treats FeF_3 and $FeCl_3$ as independent reactants, so all three species—FeF_3, $FeCl_3$, and SE—can participate simultaneously in the reaction at any mixing ratio.

4.2.2 Composition Parameterization

The interface is parameterized by the cathode molar fraction x:

$$c(x) = x \cdot \tfrac{1}{2}\,\text{FeF}_3 + x \cdot \tfrac{1}{2}\,\text{FeCl}_3 + (1-x)\cdot\text{SE} \tag{6}$$

which traces a line in the ternary (or higher-dimensional) composition space from pure SE at $x = 0$ to the 1:1 cathode mixture at $x = 1$.

4.2.3 Reaction Energy Formula

The grand-potential reaction energy per total atom of the mixture is:

$$\Delta E_{\text{rxn}}(x) = \Phi_{\text{hull}}[c(x)] - [\alpha_A\, \Phi_{\text{hull}}(\text{FeF}_3) + \alpha_B\, \Phi_{\text{hull}}(\text{FeCl}_3) + \alpha_{SE}\, \Phi_{\text{hull}}(\text{SE})] \tag{7}$$

where $\alpha_i = N^{(i)}_{\text{atoms}}/N^{\text{total}}_{\text{atoms}}$ is the atom fraction of component i in the mixture (counting all atoms including Li).

4.2.4 Grand-Potential Phase Diagram and the Li Projection

The GPCD is built in the *Li-free* composition subspace: Li is the open element, so pymatgen removes it from all entries and builds the hull over the remaining elements (e.g., {Fe, F, Cl, Y} for the FeF_3/Li_3YCl_6 system). This means:

1. The GPCD's `get_hull_energy_per_atom(comp)` method *does not accept Li-containing compositions directly.* Passing Li_3YCl_6 raises a `ValueError` because Li is not in the diagram's element list.
2. All compositions must have Li stripped before querying the GPCD.
3. The hull energy returned is *per non-Li atom.*

Normalization fix. To keep all energies on a consistent "per total atom (including Li)" basis, the following rescaling is applied:

$$\Phi^{\text{per total}}_{\text{hull}}(c) = \Phi^{\text{per non-Li}}_{\text{hull}}(\tilde{c}) \cdot \frac{N_{\text{non-Li}}}{N_{\text{total}}} \tag{8}$$

where $\tilde{c}$ is the composition with Li removed and $N_{\text{non-Li}}$, N_{total} are the corresponding atom counts.

Why this is self-consistent. Consider Li_3YCl_6 ($N_{\text{total}} = 10$, $N_{\text{non-Li}} = 7$, $\tilde{c} = YCl_6$):

$$\Phi^{\text{per total}}_{\text{hull}}(\text{Li}_3\text{YCl}_6) = \Phi^{\text{per non-Li}}_{\text{hull}}(\text{YCl}_6) \cdot \tfrac{7}{10} = \frac{E(\text{Li}_3\text{YCl}_6) - 3\mu_{\text{Li}}}{10}$$

which is exactly the grand-potential energy per total formula-unit atom. At $x = 0$ (pure SE): $\Phi^{\text{per total}}_{\text{hull}}(c_{\text{SE}}) = \Phi^{\text{per total}}_{\text{hull}}(\text{SE})$, so the reference and product energies cancel and $\Delta E_{\text{rxn}}(0) = 0$ as required. For FeF_3 and $FeCl_3$ (no Li): $N_{\text{non-Li}} = N_{\text{total}}$, so Eq. (8) reduces to the standard GPCD query.

4.2.5 Dense Scan and Kink Detection

Because there is no built-in ternary `InterfacialReactivity` class, the reaction energy is computed by a dense scan:

```
xs = np.linspace(0, 1, N_SCAN=300)  # molar fraction grid
for x in xs:
    comp_mix = x/2*FeF3 + x/2*FeCl3 + (1-x)*SE  # element-wise
    e_hull   = gpd_hull_per_total_atom(comp_mix, gpd)
    e_ref    = (a_A*e_A + a_B*e_B + a_SE*e_SE) / total_atoms
    E_rxn[x] = e_hull - e_ref
```

Kink positions (phase boundaries) are then detected by finding points where the slope $|dE_{\text{rxn}}/dx|$ changes abruptly (threshold `SLOPE_TOL`$=10^{-3}$ eV/atom per molar fraction unit).

4.2.6 Results

Outputs: `chemical_ternary_all.png`, `electrochemical_ternary_all.png`, `ternary_table.csv` (1346 rows).

Key differences from Option A:

- **Multiple kinks** appear between $x \approx 0.73$ and $x = 1.0$ that are absent in Option A. These reflect a richer ternary phase boundary structure when FeF_3 and $FeCl_3$ are allowed to react independently and with each other.
- **Smaller magnitude** near the cathode-rich end ($\Delta E_{\text{rxn}} \approx -0.03$ to $-0.04\,\text{eV/atom}$ at $x \approx 0.73$, $V = 1\,\text{V}$ for Li_3YCl_6) compared to Option A's single kink value of $-0.84\,\text{eV/atom}$ at $x = 0.57$. This is because at large x (little SE present) the reaction is mainly between the small amount of SE and the large excess of cathode—a dilute perturbation.
- The dense scan produces continuous curves with visible slope changes (kinks), whereas Option A produces a piecewise-linear curve with exactly the kink points returned by `get_kinks()`.
- At $x = 1$ (pure cathode, no SE), ΔE_{rxn} may be slightly non-zero, reflecting any thermodynamic driving force for FeF_3 and $FeCl_3$ to react with each other (form mixed Fe–F–Cl phases). This quantity is absent in Option A by construction.

4.2.7 Comparison of Option A vs. Option C

	Option A	**Option C**
Cathode representation	Single pseudo-compound $Fe_2F_3Cl_3$	Two separate phases FeF_3 + $FeCl_3$
Reference energy (cathode)	Hull energy of $Fe_2F_3Cl_3$	$\frac{1}{2}E_{\text{hull}}(FeF_3) + \frac{1}{2}E_{\text{hull}}(FeCl_3)$
Composition space	Binary (cathode vs. SE)	Ternary (FeF_3, $FeCl_3$, SE)
Kinks found	1–2 per voltage	Multiple per voltage
$FeF_3 \leftrightarrow FeCl_3$ reaction	Absorbed	Explicitly captured
Method used	`GrandPotentialInterfacialReactivity`	Dense scan + GPCD hull

5 Output Files Summary

File	Row	Description
`chemical_stability_FeF3.png`	3	Chemical, all 5 SEs
`electrochemical_stability_FeF3.png`	3	Electrochemical, 5 subplots
`stability_table_FeF3.csv`	3	652 rows, 0.1 V steps
`chemical_stability_FeCl3.png`	4	Chemical, all 5 SEs
`electrochemical_stability_FeCl3.png`	4	Electrochemical, 5 subplots
`stability_table_FeCl3.csv`	4	282 rows, 0.1 V steps
`chemical_stability_FeF3-FeCl3.png`	2A	Chemical, pseudo-compound
`electrochemical_stability_FeF3-FeCl3.png`	2A	Electrochemical, pseudo-compound
`stability_table_FeF3-FeCl3.csv`	2A	687 rows, 0.1 V steps
`chemical_ternary_all.png`	2C	Chemical, ternary method
`electrochemical_ternary_all.png`	2C	Electrochemical, ternary method
`ternary_table.csv`	2C	1346 rows, 0.1 V steps

6 Software and Data Sources

- **Database**: Materials Project (MP), accessed via `mp-api` client; database version 2025.09.25.
- **pymatgen** version: 2025.10.7 (updated API; `pd_non_grand` is now a required positional argument in `GrandPotentialInterfacialReactivity`).
- **DFT functional**: GGA (PBE) for all Fe-halide entries. No GGA+U correction is applied to Fe in halide environments within the current MP database.
- **Main scripts**:
  - `electrochemical_stability.py` — Rows 2A, 3, 4 via `InterfacialReactivity`
  - `ternary_stability.py` — Row 2C via dense GPCD scan